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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Third Semester B Tech Degree Examination December 2021 (2019 scheme)

Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

1

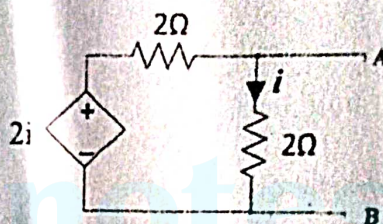
State and explain reciprocity theorem using an example

(3)

2

Determine the Norton's equivalent circuit of the following network

(3)



3

Define time constant of a circuit. Illustrate and explain does time constant affect the charging time of the capacitor in series RC circuits connected to a DC source? (3)

4

Derive the expression for the current in a series RL circuit when connected to a DC source of voltage V , at time $t = 0$. Assume zero initial conditions (3)

5

Obtain the transfer function of a typical series RLC circuit. Take the voltage across the capacitor as the output variable (3)

6

Define coefficient of coupling in coupled circuits. What are its maximum and minimum values? (3)

7

A series RLC circuit with $R = 10\Omega$, $L = 2H$ and $C = 0.5F$ is connected to a 230V, variable frequency AC source. Determine the frequency of the source at which the circuit current is maximum. Also find the maximum current. (3)

8

Describe the variation of the impedance, power factor and current as a function of frequency in a series resonant circuit. (3)

9

What are h-parameters? Why are they called hybrid parameters? (3)

10

Show that the overall T-parameter matrix of two cascaded 2-port networks is the product of the T-parameter matrix of the individual networks. (3)

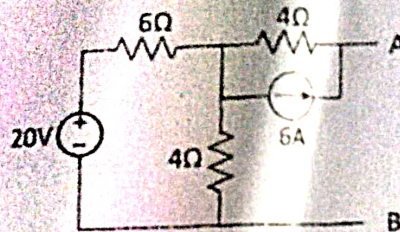
PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

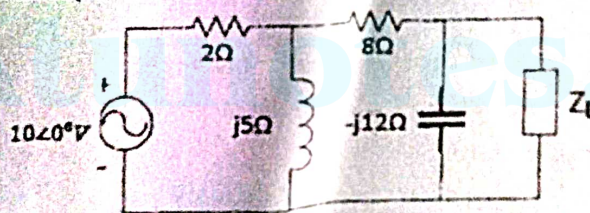
11. For the network given below,

- a) Find the Thevenin's equivalent circuit across the terminals A and B (10)
 b) Determine the power dissipated in a 10Ω resistance when it is connected across the terminals A and B (4)



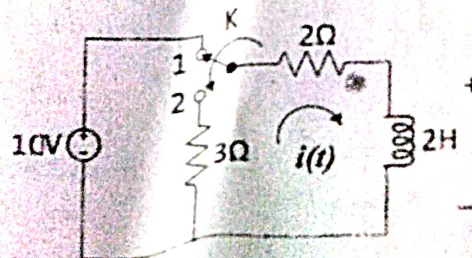
12. In the circuit given below,

- a) Determine the value of the load impedance for maximum power transferred by the source to the load (12)
 b) Find the maximum power transferred. (2)



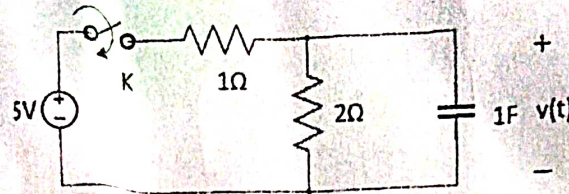
Module 2

13. a) In the circuit shown below, the switch was initially at position 1 and the steady state condition is reached. At $t = 0$, the switch is changed to position 2. Determine the expression for the current $i(t)$, for $t > 0$. (8)



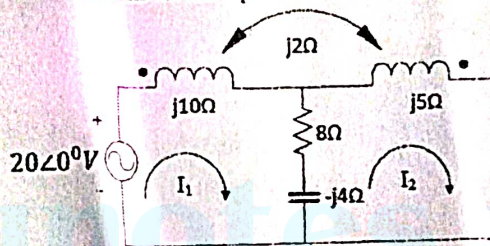
- b) A $0.25F$ capacitor with an initial voltage of $10V$ is connected across a coil of 5Ω resistance and $1H$ inductance, at time $t = 0$. Determine the expression for the current through the coil for $t > 0$. (6)

- 14 In the circuit given below, the switch K is closed at $t = 0$
- a) Determine the expression for the voltage across the capacitor, $v(t)$ for $t > 0$ (10)
- b) Calculate the value of $v(t)$ at $t = 1$ seconds and its final steady state value. (4)

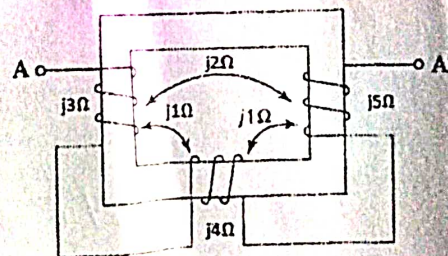


Module 3

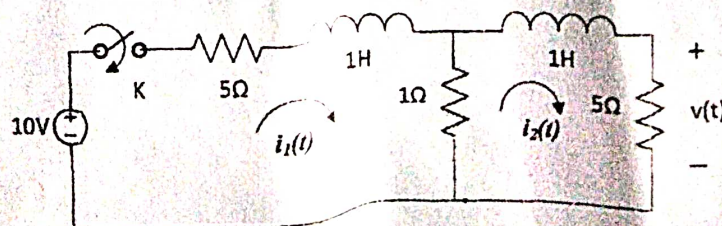
- 15 a) Obtain the conductively coupled equivalent circuit for the network given below. (7)
- Also write the mesh equations for the equivalent circuit



- b) If the network given below is connected across a 230V, 50Hz AC source, determine the current supplied by the source (7)



- 16 The switch K in the circuit given below is closed at $t = 0$.
- a) Determine the transformed circuit for $t > 0$ Assume zero initial conditions (4)
- b) Find the time domain expression for the voltage $v(t)$ across the 5Ω resistor for $t > 0$ Use mesh analysis (10)



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Module 4

17. A 3-phase, 3-wire, 240V system supplies a delta-connected load in which (14)
 $Z_{AB} = 25\angle 90^\circ \Omega$, $Z_{BC} = 15\angle 30^\circ \Omega$ and $Z_{CA} = 20\angle 0^\circ \Omega$. Determine the phase currents, line currents and total power consumed by the load
18. A 3-phase, 400V, 4-wire system has a star connected load with $Z_A = 20\Omega$, (14)
 $Z_B = 15 + j10\Omega$ and $Z_C = j5\Omega$. Find the line currents, current through the neutral conductor and the total power consumed by the load

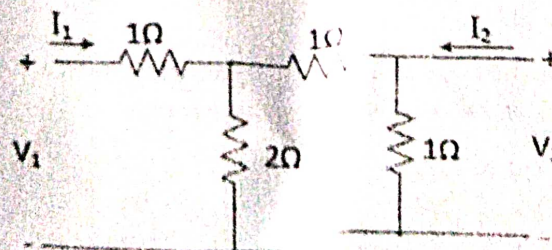
Module 5

19. a) A two port network 'A' has $[Z]_A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and another two port network 'B' (7)
has $[Y]_B = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. If the two networks A and B are connected in series, find the Z parameters of the overall network
- b) Find the equivalent π network of a two port network represented by the (7)
following equations

$$V_1 = 2I_1 + I_2$$

$$V_2 = I_1 + 3I_2$$

20. a) Find the h-parameters of the following two port network (8)



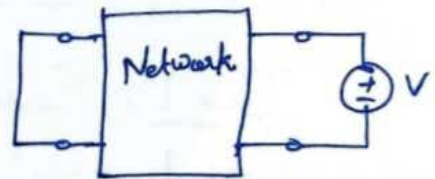
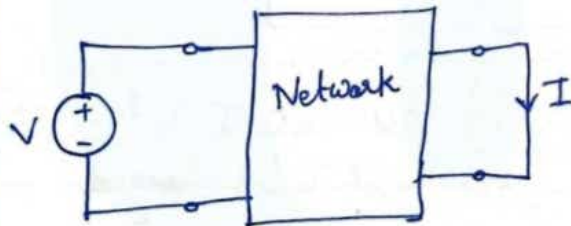
- b) Derive the conditions of symmetry and reciprocity of a two port network in terms (6)
of ABCD parameters

1

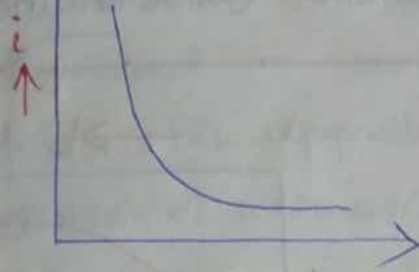
RECIPROCITY THEOREM

It states that, "in a linear, bilateral, active, single source network, the ratio of excitation to response remains same when the positions of excitation and response are interchanged".

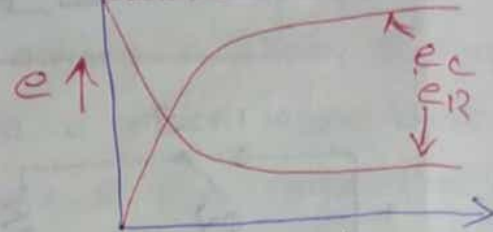
If a single voltage source V_a in the branch 'a' produces a current I_b in the branch 'b' then if the voltage source removed and inserted in the branch 'b', it will produce a current I_b in branch 'a'.



3



Variation of i with time.



Variations of e_R & e_C with time.

The transient voltages,

$$e_R = Ri = R \times I e^{-(t/\lambda)} = E \cdot e^{-(t/\lambda)} \quad \text{--- (7)}$$

But $e_R + e_C = E$

$$\therefore e_C = E - e_R = E [1 - e^{-(t/\lambda)}] \quad \text{--- (8)}$$

e_R - exponentially decreases from E to 0 as 't' increases from 0 to ∞ .

e_C - exponentially increases from 0 to E as 't' increases from 0 to ∞ .

Time Constant (λ)

In (8), sub: $\lambda = t$

$$\text{Then } i = I \times e^{-1} = 0.368 \times I$$

= 36.8% of the initial current

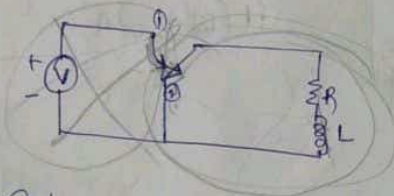
$\therefore$ Time Constant for the RC circuit is the time during which the current falls to 36.8% of initial current.

Note: e_C can also be obtained from

$$e_C = \frac{1}{C} \int i(t) dt$$

4

RL Decaying transients

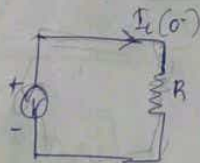


@ $t=0$

switch is in position (2).

@ $t=0^-$

switch is in position (1)



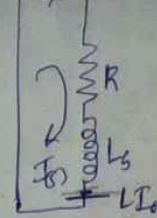
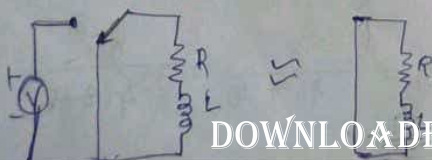
$L \rightarrow$ short.

$$I_L(0^-) = I_L(0^+) = \frac{V}{R}$$

@ $t \geq 0$

$t=0^+$

switch is in position 2.



$$\cancel{RI(s)} + LsI(s) - LI_0 = 0$$

$I_0 =$

$$RI(s) + LsI(s) - LI_0 = 0$$

$$I(s) = \frac{LI_0}{R+Ls} = \frac{LI_0}{L\left(\frac{R}{L}+s\right)}$$

$$\frac{I_0}{s + R/L} \rightarrow I(t) = I_0 e^{-R/L t}$$

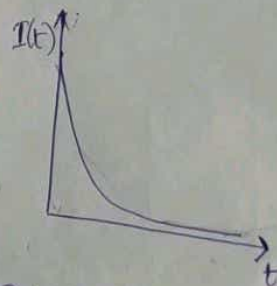
$$I(t) = \frac{V}{R} e^{-R/L t}$$

@ $t=0$

$$I(t) = \frac{V}{R}$$

@ $t=\infty$

$$I(t) = 0$$

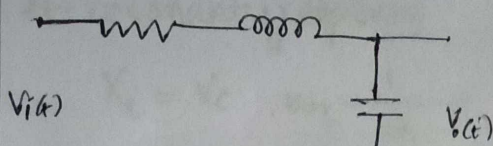


@ $t = R/L$

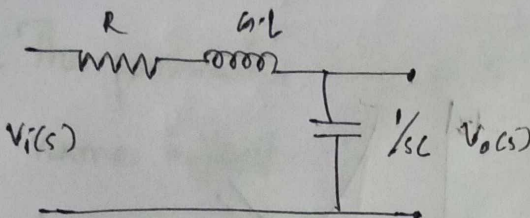
$$I(t) = \frac{V}{R} 0.36$$

here time constant is time take for the current to decay to its final 36%

h)



iii)



$$V_{out} = \frac{V_i(s) \times \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}}$$

$$H(s) = \frac{1}{1 + R s C + s^2 L C}$$

6) Coefficient of coupling.

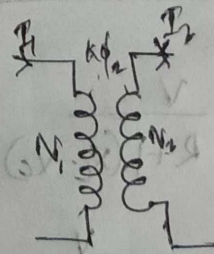
Fraction of magnetic flux produced by the current in one coil that links with a nearby coil

$$\phi = \frac{N_1 \phi_1}{I_1}$$

$$M_{12} = \frac{K \phi_1 \times N_2}{I_1}$$

$$L_2 = \frac{\phi_2 N_2}{I_2} \quad M_{21} = \frac{K \phi_2 N_1}{I_2}$$

$$M_{12} = M_{21} = M$$



$$k = \frac{M}{\sqrt{L_1 L_2}}$$

The coefficient of coupling is always less than unity and has a maximum value of 1

8 Impedance at Resonance

$$Z = R + j(X_L - X_C)$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right)$$

At resonance $\omega L = \frac{1}{\omega C}$

$$\therefore Z = R$$

The variation of Z with frequency is shown.

the magnitude of Z is minimum at resonant

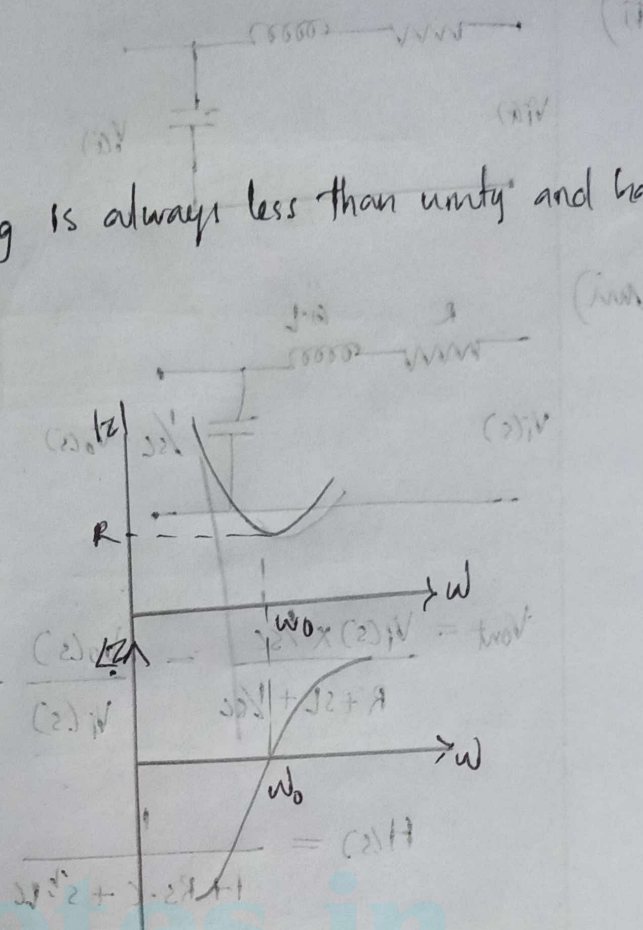
frequency and equal to R

Current at resonance.

$$I = \frac{V}{R + j(X_L - X_C)}$$

$$= \frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$|I| = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$



at resonant frequency

$$X_L = X_C \quad \omega_L = \frac{1}{\omega_C}$$

$$F = \frac{V}{R}$$

9

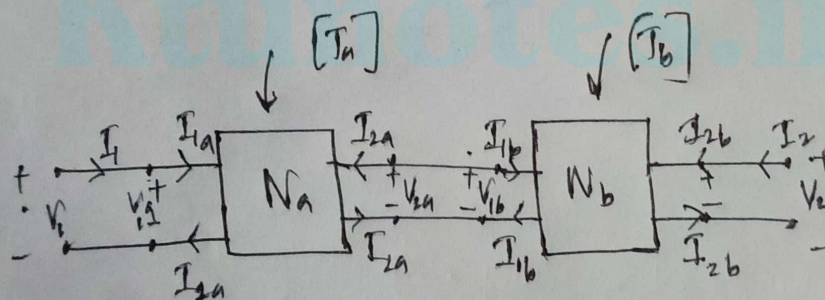
The parameters have a mixture of units as Ω and Ω^{-1} , so the name hybrid.

here the voltages of one port and current of other port are taken as the independent variables.

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

10



$$V_1 = V_{1a}, \quad I_1 = I_{1a}, \quad V_{2a} = V_{1b}, \quad I_{2a} = I_{1b}; \quad V_{2b} = V_2 \text{ and } I_{2b} = I_2$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix}$$

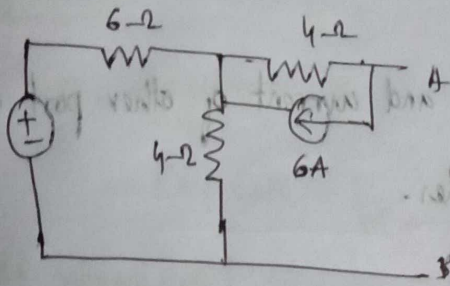
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = [T_a] \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix}$$

$$\begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix} = [T_b] \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

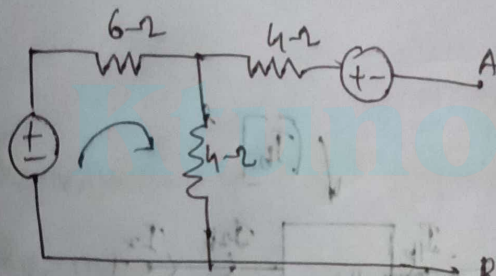
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = [T_a] [T_b] \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \underbrace{[T_a] \times [T_b]}_{[T]} \times \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

$$[T] = [T_a] \times [T_b]$$



$$6 \times 4 = 24V$$

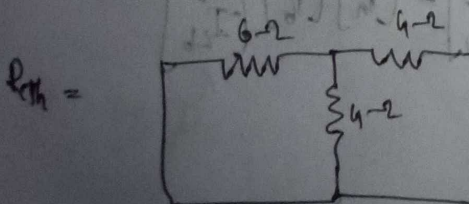


$$\frac{V_{oc} - 20}{6} + \frac{V_{oc}}{4} + \frac{V_{oc} + 24}{4} = 0$$

$$\frac{V_{oc}}{6} + \frac{V_{oc}}{4} + \frac{V_{oc}}{4} = \frac{109}{12}$$

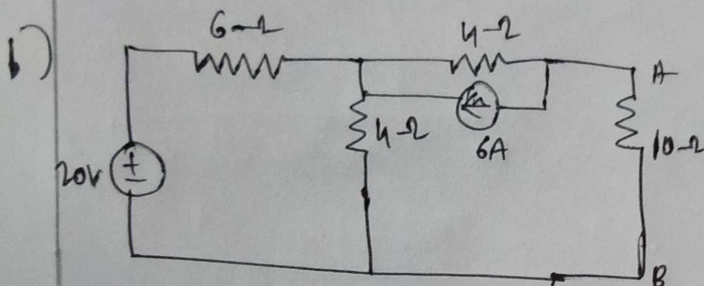
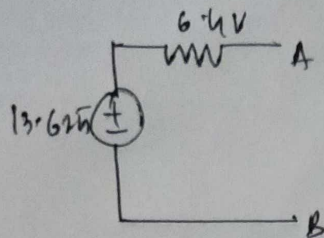
$$\frac{2}{3} V_{oc} = \frac{109}{12} \Rightarrow \underline{13.625V}$$

$$V_{oc} = 13.625V$$



$$6 \parallel 4 + 4 = 6.4\Omega$$

Thevenin's equivalent circuit.



$$P = I^2 R$$

Current through 10Ω resistor $= I_L$

$$I_L = \frac{V_{OC}}{R_{th} + R_L} = \frac{13.62V}{6.4 + 10}$$

$$= \underline{\underline{0.8307A}}$$

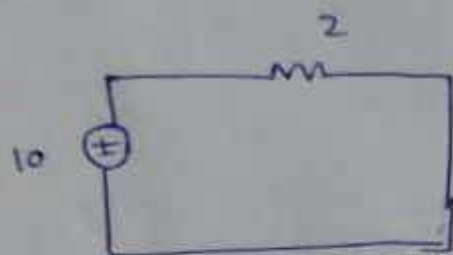
$$P = I^2 R$$

$$= (0.8307)^2 \times 10$$

$$= \underline{\underline{6.902W}}$$

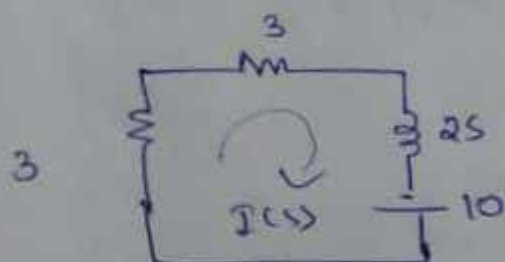
13
(a)

when switch at postⁿ (1)



$$I_L(0^-) = \frac{10}{2} = 5 \text{ A}$$

at postⁿ (2)



$$V_L = L I_C$$

$$2 \times 5$$

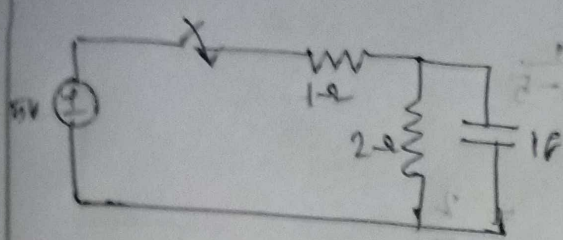
$$-I_1(s) 3 - 3I_1(s) - 2S I_1(s) + 10$$

$$I_1(s) [3 + 3 + 2S] = 10$$

$$I_1(s) = \frac{10}{6 + 2S}$$

$$I_1(s) = \frac{10}{2(S + 3)} = \frac{5}{S + 3}$$

$$I_1(t) = \underline{\underline{5 \cdot e^{-3t}}}$$

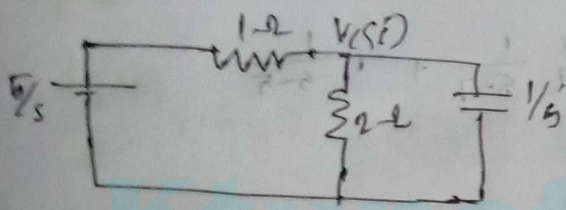


switch closed at $t=0$

at $t=0^-$ switch opened

$$V_C(0^-) = 0 = V_C(0^+)$$

At $t>0$ switch is closed



$$\frac{V_C(s) - 5/s}{1} + \frac{V_C(s)}{2} + \frac{V_C(s)}{1/s} = 0$$

$$V_C(s) \left[1 + \frac{1}{2} + s \right] = 5/s$$

$$V_C(s) [1.5 + s] = 5/s$$

$$V_C(s) = \frac{5}{s(1.5 + s)}$$

$$= \frac{A}{s} + \frac{B}{s + 1.5}$$

$$A = s V_C(s) \text{ at } s=0$$

$$\frac{5}{s(1.5 + s)} = \frac{5}{1.5}$$

$$b = \frac{5}{5(1.5+s)} (1.5+s) \quad \text{at } s = -1.5$$

$$= \frac{-5}{1.5}$$

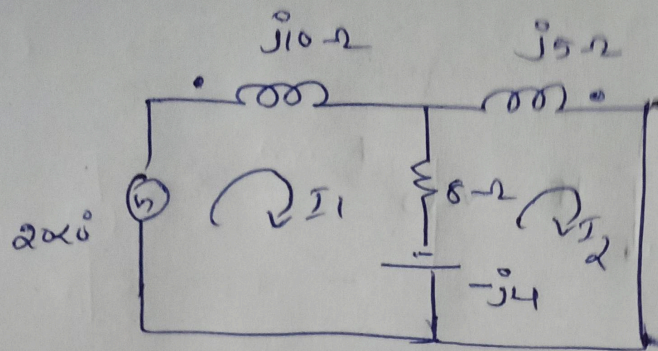
$$V_1(s) = \frac{1.5}{1.5s} + \frac{-5}{1.5(s+1.5)}$$

$$V_c(t) = \frac{5}{1.5} - \frac{5}{1.5} e^{-1.5t}$$

$$V_c(t) \text{ at } t=1 = \frac{5}{1.5} - \frac{5}{1.5} e^{-1.5 \times 1}$$

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15 (a)



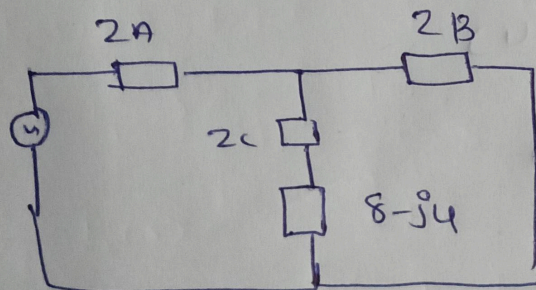
$$X_m = 12\Omega$$

m is -ve

$$Z_A = L_1 - M ; Z_B = L_2 - M ; Z_C = M$$

$$Z_A = j10 - j12 = -j2\Omega \quad Z_C = j12\Omega$$

$$Z_B = j5 - j12 = -j7\Omega$$



Conductively coupled circuit

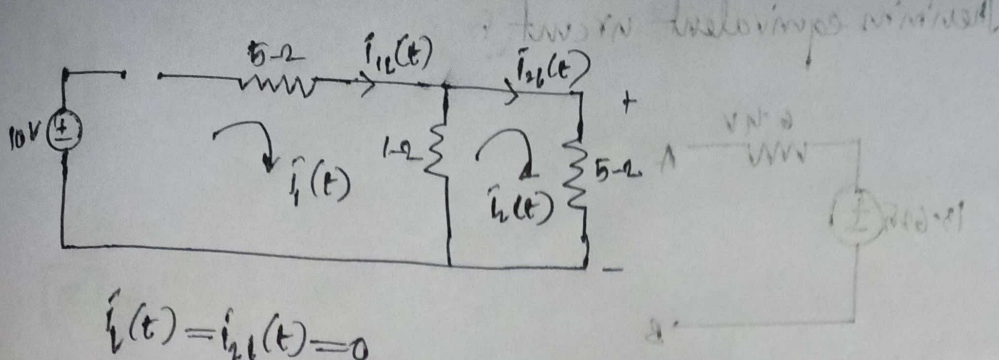
$$2\angle 0^\circ - I_1(j10) + (I_1 - I_2)(8 - j4) + j12 \quad \text{--- (1)}$$

$$- I_2(j5) - (I_2 - I_1)(8 - j4) \quad \text{--- (2)}$$

mesh

16.

1)



2)

loop ①

$$5i_1(t) + \frac{di_1(t)}{dt} + 1[i_1(t) - i_2(t)] = 10u(t) \quad \text{--- (1)}$$

loop ②

$$\frac{di_2(t)}{dt} + 5i_2(t) + 1[i_2(t) - i_1(t)] = 0 \quad \text{--- (2)}$$

taking laplace transform.

$$5I_1(s) + sI_1(s) + I_1(s) - I_2(s) = \frac{10}{s} \quad \text{--- (3)}$$

$$sI_2(s) + 5I_2(s) + I_2(s) - I_1(s) = 0 \quad \text{--- (4)}$$

$$(3) \rightarrow I_1(s)[6+s] - I_2(s) = \frac{10}{s} \quad \text{--- (5)}$$

$$(4) \rightarrow -I_1(s) + I_2(s)[6+s] = 0 \quad \text{--- (6)}$$

$$I_1(s) = I_2(s)[6+s] \quad \text{--- (6)}$$

sub (6) in (5)

$$I_2(s)(6+s)^2 - I_2(s) = \frac{10}{s}$$

$$I_2(s) = \frac{10}{s}$$

$$= \frac{k_1}{s} + \frac{k_2}{s-7} + \frac{k_3}{s-5}$$

$$\text{where } k_1 = \frac{10}{(s-7)(s-5)} \Big|_{s=0} = \frac{2}{7}$$

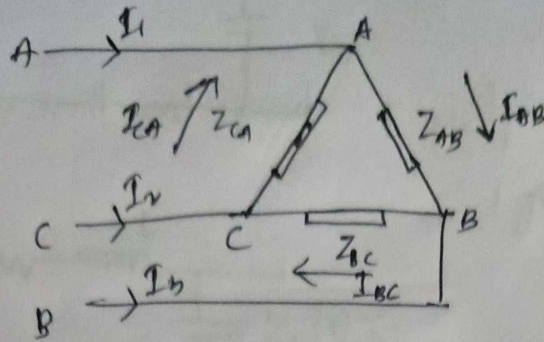
$$k_2 = \frac{10}{s(s-5)} \Big|_{s=7} = \frac{5}{7}$$

$$k_3 = \frac{10}{s(s-7)} \Big|_{s=5} = \underline{\underline{-1}}$$

$$\therefore I_2(s) = \frac{2}{7} \frac{1}{s} + \frac{5}{7} \frac{1}{s-7} + -1 \frac{1}{s-5}$$

$$\begin{aligned} I_2(t) &= \mathcal{L}^{-1}[I_2(s)] \\ &= \left(\frac{2}{7} + \frac{5}{7} e^{7t} - e^{-5t} \right) u \end{aligned}$$

$$V(t) = 5 \left(\frac{2}{7} + \frac{5}{7} e^{7t} - e^{-5t} \right) u$$



$$V_{AB} = 240 \angle 0^\circ \text{ V}$$

$$V_{BC} = 240 \angle -120^\circ \text{ V}$$

$$V_{CA} = 240 \angle 120^\circ \text{ V}$$

$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{240 \angle 0^\circ}{25 \angle 90^\circ} = \underline{\underline{9.6 \angle -90^\circ \text{ A}}}$$

$$I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{240 \angle -120^\circ}{15 \angle 30^\circ} = \underline{\underline{16 \angle -150^\circ \text{ A}}}$$

$$I_{CA} = \frac{V_{CA}}{Z_{CA}} = \frac{240 \angle 120^\circ}{20 \angle 0^\circ} = \underline{\underline{12 \angle 120^\circ \text{ A}}}$$

Line current

$$I_1 = \frac{V_{AB}}{Z_1}$$

$$I_1 = I_{AB} - I_{CA}$$

$$= (9.6 \angle -90^\circ) - (12 \angle 120^\circ)$$

$$\underline{\underline{20.78 \angle 73.22^\circ}}$$

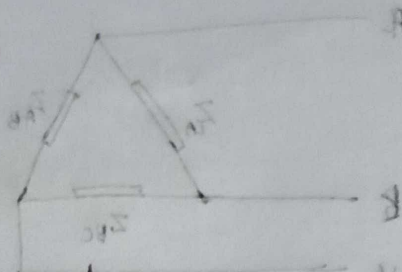
$$I_2 = I_{BC} - I_{AB}$$

$$= (16 \angle -150^\circ) - (9.6 \angle -90^\circ)$$

$$I_B = I_{CA} - I_{BC}$$

$$12 \angle 120^\circ - 16 \angle -150^\circ$$

$$= 10.91 \angle 66.78^\circ$$



To calculate the power consumed.

$$\text{Power in phase 1} = (I_{AB})^2 \times R_1$$

$$= (9.6)^2 \times 2$$

$$Z_1 = 25 \angle 90^\circ$$

$$= 25j$$

$$Z_2 = 15 \angle 30^\circ = 12.9 + 7.5j$$

$$Z_3 = 20 \angle 0^\circ = 20$$

$$\text{Power in phase 2} = (I_{BC})^2 \times R_2$$

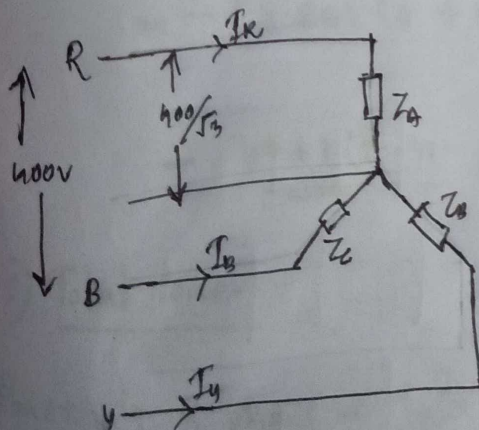
$$16^2 \times 12.9$$

$$\text{Power in phase 3} = (I_{CA})^2 \times R_3$$

$$12^2 \times 20$$

$$\text{Total power in load} = 92.16 + 3302.4 + 2880 \text{ W}$$

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$$Z_A = 20 \Omega$$

$$Z_B = 15 + j10 \Omega$$

$$Z_C = j5 \Omega$$

$$V_L = 400 \text{ V}$$

$$V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$$V_{RN} = 230.94 \angle 0^\circ \text{ V}$$

$$V_{YN} = 230.94 \angle -120^\circ \text{ V}$$

$$V_{BN} = 230.94 \angle 120^\circ \text{ V}$$

a) Line currents.

$$I_R = \frac{V_{RN}}{Z_A} = \frac{230.94 \angle 0^\circ}{20}$$

$$= \underline{11.547 \angle 0^\circ \text{ A}}$$

$$I_Y = \frac{V_{YN}}{Z_B} = \frac{230.94 \angle -120^\circ}{15 + j10}$$

$$= \underline{12.7 \angle -153.9^\circ \text{ A}}$$

$$I_B = \frac{V_{BN}}{Z_C} = \frac{230.94 \angle -240^\circ}{j5}$$

$$= \underline{46.17 \angle 30^\circ \text{ A}}$$

b) Neutral current.

$$I_N = -(I_R + I_Y + I_B)$$

$$= -(11.547 \angle 0^\circ + 12.7 \angle -153.9^\circ + 46.17 \angle 30^\circ)$$

$$= \underline{43.68 \angle 23.31^\circ \text{ A}}$$

c) Total power

$$\text{Power in R phase} = I_R^2 \times R$$

$$= (11.55)^2 \times 20$$

$$\text{Power in Y phase} = I_Y^2 \times R_Y \\ = (12.7)^2 \times 10 \text{ mW} =$$

$$\text{Power in B phase} = I_B^2 \times R_B \\ = (46.17)^2 \times 0$$

$$\text{Power absorbed by the load} = \text{Power in (R phase + Y phase + B phase)}$$

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a) $[Z]_A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$$[Y]_B = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$[Z]_B = [Y]_B^{-1}$$

$$\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$Z_{\text{parameter of overall network}} = [Z]_A + [Z]_B$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\underline{\underline{\begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}}}$$

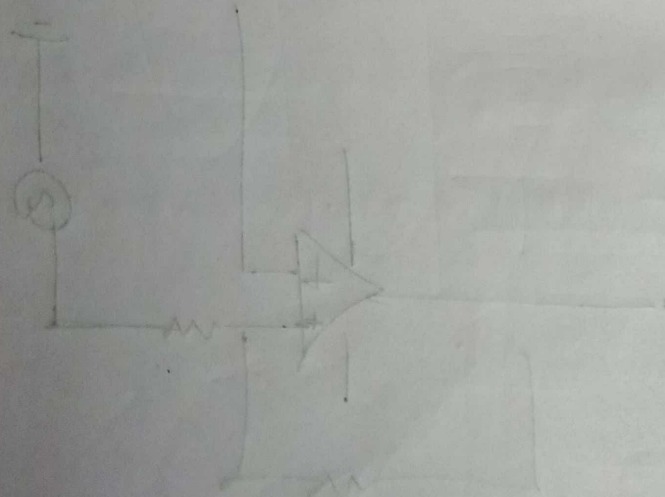
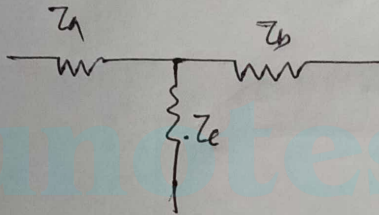
$$b \quad V_1 = 2I_1 + I_2$$

$$V_2 = I_1 + 3I_2$$

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

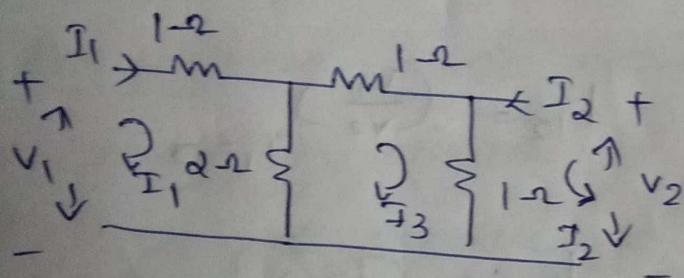
$$= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$Z_c = 1 \quad Z_a + Z_c = 2 \quad Z_b + Z_c = 3$$



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$V_2 = h_{21} I_1 + h_{22} V_2$$



$$V_1 = I_1 + 2(I_1 - I_3)$$

$$V_1 = \frac{I_1 + 2I_1 - 2I_3}{3I_1 - 2I_3} \quad \text{--- (1)}$$

$$V_2 = I_2 + 3I_3 \quad \text{--- (2)}$$

$$2(I_3 - I_1) + 2I_3 + 1(I_3 + I_2) = 0$$

$$2I_3 - 2I_1 + 2I_3 + I_3 + I_2 = 0$$

$$6I_3 - 2I_1 + I_2 = 0$$

$$6I_3 = 2I_1 - I_2$$

$$I_3 = \frac{I_1}{3} - \frac{I_2}{6}$$

$$V_1 = 3I_1 - 2\left(\frac{I_1}{3} - \frac{I_2}{6}\right)$$

$$V_2 = I_2 + 3\left[\frac{I_1}{3} - \frac{I_2}{6}\right]$$

$$V_1 = 3I_1 - \frac{2I_1}{3} + \frac{I_2}{3}$$

$$V_1 = \frac{9I_1 - 2I_1}{3} + \frac{I_2}{3} \Rightarrow \frac{7I_1}{3} + \frac{I_2}{3} \quad \text{--- (1)}$$

$$V_2 = I_2 + I_1 - \frac{I_2}{3}$$

$$V_2 = \frac{3I_2 - I_2}{3} + I_1 = \frac{2I_2}{3} + I_1 \quad \text{--- (2)}$$

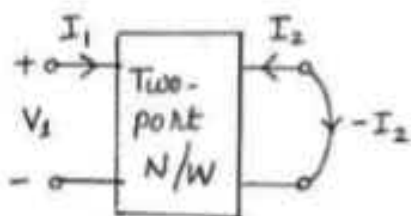
$$3V_1 = 7I_1 + I_2$$

$$V_2 = \frac{2I_2}{3} + I_1$$

Rearranging the equation we get the h-parameters

III. T-parameters.

(a) Reciprocity in T-parameter representation.



We have,

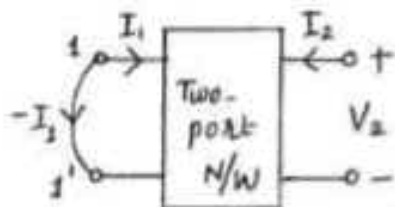
$$\begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned} \quad \text{--- (1)}$$

considering the SC condition ($V_2 = 0$) and the actual current direction through 2, 2',

$$\begin{aligned} \text{(1)} \rightarrow V_1 &= -BI_2 \\ I_1 &= -DI_2 \end{aligned} \quad \text{--- (2)}$$

From (2), we get $\frac{I_1}{V_1} = -\frac{1}{B}$ --- (3)

After interchanging the position of excitation and response,



In (1), considering the SC condition ($V_2 = 0$) and the actual current direction through 1, 1',

$$\begin{aligned} \text{(1)} \rightarrow 0 &= AV_1 - BI_1 \\ I_1 &= CV_1 - DI_1 \end{aligned} \quad \text{--- (4)}$$

From (4), $AV_2 = BI_2$ & $I_2 = \frac{A}{B}V_2$

$$\begin{aligned} I_1 &= CV_2 - DI_2 = CV_2 - D\left[\frac{A}{B}V_2\right] \\ &= CV_2 - \frac{AD}{B}V_2 \\ &= \left[C - \frac{AD}{B}\right]V_2 \end{aligned}$$

$$\frac{I_1}{V_2} = C - \frac{AD}{B} \quad \text{--- (5)}$$

Assuming $V_1 = V_2$ (same source is connected across the port terminals), from (3) & (5),

$$C - \frac{AD}{B} = -\frac{1}{B} \quad \text{u,} \quad BC - AD = -1$$

Thus, $AD - BC = 1$ or $\begin{vmatrix} A & B \\ C & D \end{vmatrix} = 1$

represents the condition of reciprocity.

(b) Symmetry in T-parameter representation.

We have $Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} = \frac{AV_2 - BI_2}{CV_2 - DI_2} \bigg|_{I_2=0}$

$$= \frac{A}{C}$$

Also, $Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0} = \frac{D}{C}$

$$\because I_1 = CV_2 - DI_2$$

$$0 = CV_2 - DI_2 \Rightarrow CV_2 = DI_2 \Rightarrow \frac{V_2}{I_2} = \frac{D}{C}$$

For symmetrical Z-parameter network,

$$\begin{aligned} Z_{11} &= Z_{22} \\ \Rightarrow \frac{A}{C} &= \frac{D}{C} \Rightarrow \underline{A = D} \end{aligned}$$

Summary

(1) Condition for reciprocity : $AD - BC = 1$

(2) Condition for symmetry : $A = D$